NOVA IMS
INFORMATION MANAGEMENT SCHOOL

AMAZING CREATIVE DESK
COWORKING

BEYOND THE PLAN: UNCOVERING WHO ACDC'S MEMBERS REALLY ARE

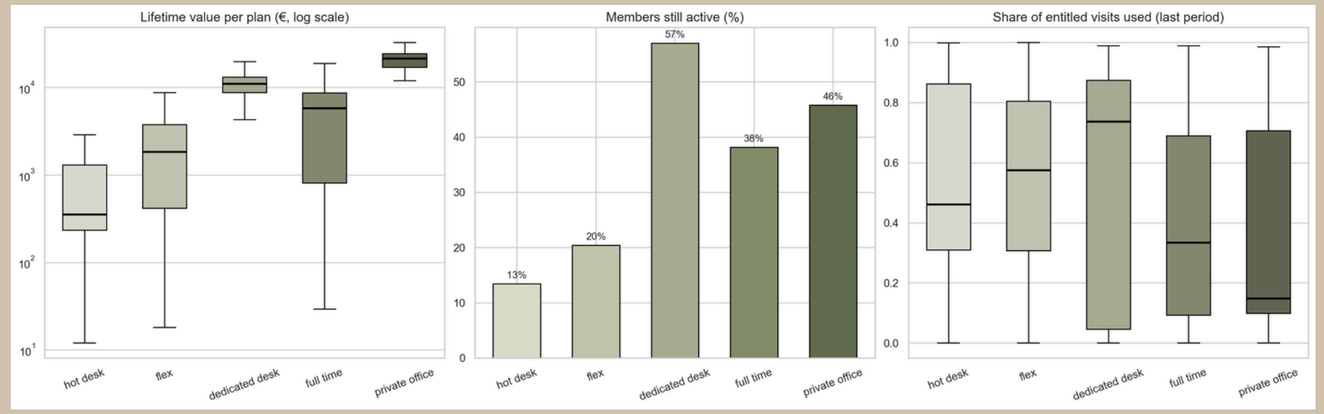
DATA MINING
EXPLORATORY DATA ANALYSIS

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WHAT SURPRISED US

Unexpected findings and what they say about our future clusters

PAYING MORE ≠ USING MORE



IMPLICATION FOR CLUSTERING

Value and usage are separate dimensions. We will segment on both, because the meaningful differences are within plans, not between them.

- **Dedicated desk outlasts Full Time**: Both give 5 days a week, yet Dedicated Desk has about 2 times the lifetime value and 57% of its members are still active, against 38%.
- **Private Office pays the most but rarely shows up**: It has the highest value but uses only 15% of its entitled days. It looks more like a business address than a workspace.
- **Retention is ACDC's core issue**: Only 26% of members are active and 50% never renewed. 1 in 3 active members haven't visited in 90+ days.

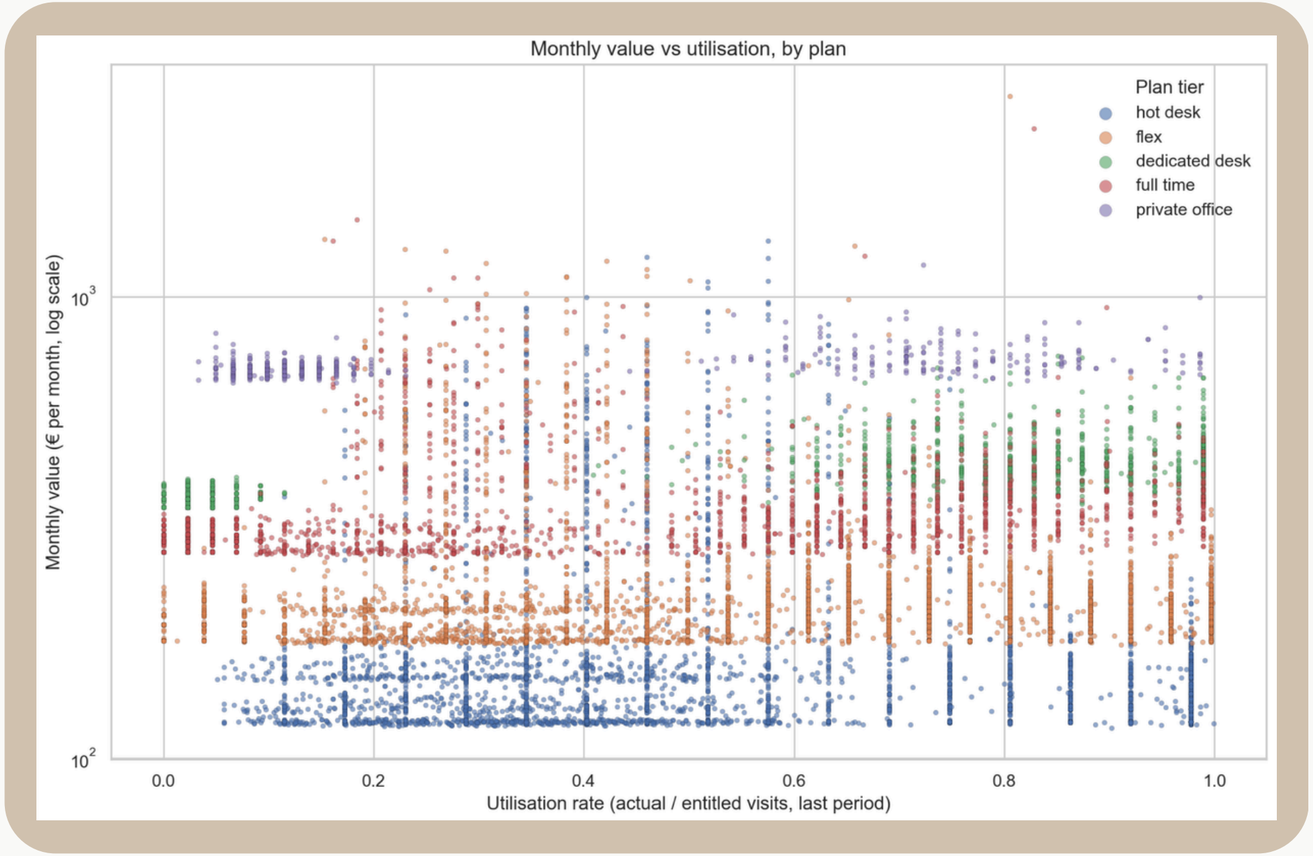
DATA RISKS AND HOW WE HANDLE THEM

The data limitations that could distort clustering, and our fixes

What we found	What we did or plan for preprocessing
1280 members with “999” days since their last visit, 562 with -1 colleagues, 143 aged under 18 or over 100, 96 on plans that don’t exist	Turned them into missing values
77 lifetime values recorded in cents and 19 meeting-room values in minutes	Converted back to euros and hours
182 duplicate rows, 149 memberships that end before they start, 88 badge swipes after the membership ended or after the data was extracted	Removed them: 419 rows, 2.7% of the data
253 members marked as not using a service they had hours for	Fixed the flags to match the hours
After cleaning, lifetime value is missing for 18% of members	Delivery 2: fill it from related columns where we can, and within each plan otherwise
Most members never use meeting rooms, events or guest passes, but a few use them a lot	Delivery 2: log scale or yes/no flags, so a few heavy users don’t pull the clusters
Some columns repeat each other (weekly visits vs last-period visits, $\rho = 0.94$), and the plan sets most of the monthly value ($\rho = 0.80$)	Flagged for now. In Delivery 2 we keep one of each pair and cluster value and behaviour separately
Around 300 unusual members found by each outlier method, almost none by both	Kept and flagged: most are our best customers, not errors

NATURAL GROUPING IN MEMBER BEHAVIOUR

Activity patterns and engineered features that point to distinct segments



- **Usage splits the premium plans in two**: On , 35% of members use less than 20% of their days while 40% use more than 80%. On Private Office, 60% use less than 20%.
- **Add-ons show how people work**: Private Office members use mail handling (77%) and phone booths (70%), like a business base. Dedicated Desk members keep things in storage (46%). Hot Desk and Flex members top up with day passes.
- **Age, team size and industry barely change the plan mix**: We'll use them to describe the clusters, not to build them.

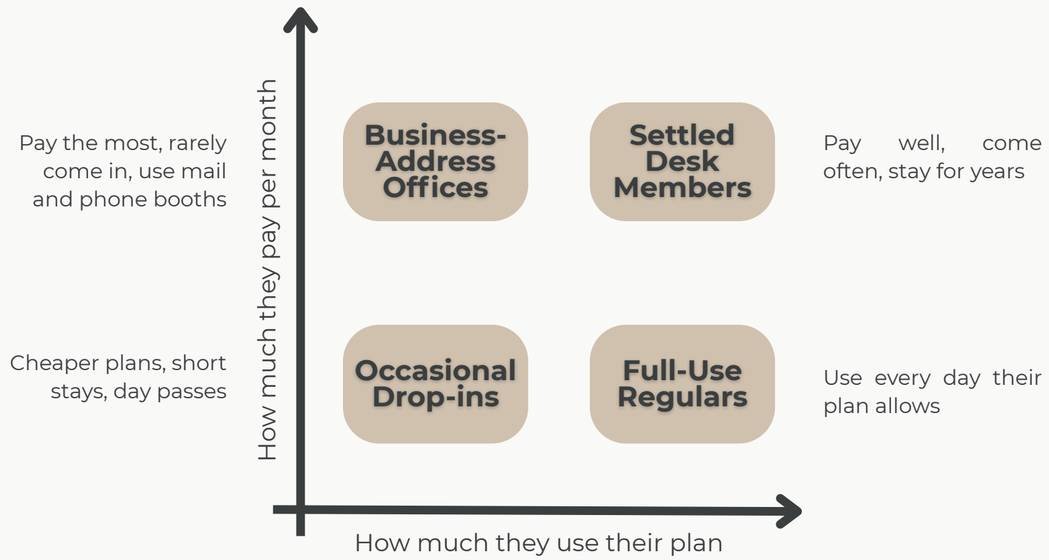
The patterns in the previous slide only appear because of these features. Raw lifetime value mostly reflects how long someone stayed, and raw visits depend on the plan, so we turned them into rates that compare every member on equal terms.

Feature	How	What it shows
Revenue per month	Lifetime value ÷ months as a member	Lifetime value mostly measures how long someone stayed (correlation of 0.95 with tenure). Per month, Dedicated Desk is worth more than Full Time: 407 euros against 313
Utilisation rate	Actual ÷ entitled visits	Every plan on the same scale, from 0 to 1. Private Office members use 15% of their days and Dedicated Desk members 74%
Membership duration	Days ÷ 30.44	Hot Desk members stay about 3 months, while Dedicated Desk and Private Office members stay more than 2 years
Renewals per year	Renewals ÷ years as a member	Loyalty that can be compared between members: 0 on Hot Desk, 0.7 on Full Time and 1 on Dedicated Desk
Plan tier level	Plans ranked from 1 (Hot Desk) to 5 (Private Office)	Lets the clustering read the plan as a number. It overlaps with revenue per month (correlation of 0.80), so we will not use both
Total add-ons used	Number of the 9 services each member uses	Hot Desk and Flex members use 1 service, Dedicated Desk and Private Office members use 3
Service hours per month	Meeting room and event space hours divided by the months as a member	Most members never book these spaces, but a few book them heavily. It needs a log scale before clustering
Guest passes per month	Guest passes divided by the months as a member	Three quarters of members never bring guests. The few who do may be a client-facing group
Total account members	Colleagues on the account plus the member	Most accounts are a single person. It barely separates members, so it will describe the clusters rather than build them

SEGMENTATION IN PLAIN TERMS

How many groups we expect, what sets them apart, and what could go wrong

We expect 4 or 5 member groups



What could get in the way:

The plan could take over and simply repeat the price list. We will look at value and behaviour separately.

A few very heavy users could pull the groups. We will use a log scale instead of removing them.

Revenue is missing for 18% of members. We will estimate it from the plan and the time as a member.

What tells them apart?

1. How much do they pay per month?
2. How much of their plan do they actually use?
3. How long do they stay, and do they renew? And which extras they use?